

NetFlowTotal: A Cloud Service Integration Platform for Malicious Traffic Analysis and Collaboration

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ABSTRACT

Network security lacks the verification of real world net flow data, and lacks a platform to collect and integrate net flow data and threat intelligence, so as to generate an evaluation benchmark for machine learning on cybersecurity. NetFlowTotal develop many net flow analysis tools to detect malicious threats in the net flow data. Through the two-side market strategies, NetFlowTotal platform tie together two distinct groups of users in a network. One kind of user can upload net flow data to the NetFlowTotal platform to obtain security incidents reports; the other kind of user can share threat intelligence to the NetFlowTotal platform to obtain more associate threat intelligence according to global net flow data. The goal of this paper is to establish a net flow evaluation platform to provide real world dataset with security incidents reports for machine learning evaluation.

CCS Concepts

• Security and privacy → Intrusion/anomaly detection and malware mitigation → Intrusion detection systems → Artificial immune systems.

Keywords

Cyber Threat Intelligence; MapReduce; Serverless; Microservice.

1. INTRODUCTION

In recent years, with the rapid and continuous change of cyber-attack methods, enterprises are facing more and more serious cyber threats and challenges. Cyber criminals can conduct cyber-attacks against specific organizations, industries and customers. It is becoming increasingly difficult for a single company to compete with an organized and specialized hacking group. Faced with Cyber threats from all over the world, enterprises have also realized that they need to have powerful Cyber Threat Intelligence (CTI) services to help them make the most correct security decisions. In addition, the inter-industry threat information sharing mechanism can effectively provide a virtuous circle of early-stage security information warning, joint defense and

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ICCB 2019, October 18–20, 2019, TAICHUNG, Taiwan

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ACM ISBN 978-1-4503-7290-9/19/10...\$15.00

<https://doi.org/10.1145/3366650.3366669>

security protection improvement, and reduce the risk of security incidents and forced interruption of business operations.

According to 2019 SANS CTI survey [1], the greatest percentage of respondents (66%) took advantage of open source or public CTI threat feeds, followed closely by threat feeds from security vendors (64%), community or industry groups (63%), external sources such as media and news (63%), and security data such as alerts gathered from the IDS, firewall, endpoints and other security systems (62%). Although the choices were very close in their distribution, it still seems that external sources are often favored over internal sources for gathering CTI. This could be a result of the ease of obtaining external sources compared with how difficult it can be to get the right collection within the organization. It is worth mentioning that the tenth most popular was network traffic analysis packet and flow at 53%, so this paper establishes a net flow evaluation platform to provide real world dataset with security incidents reports for machine learning evaluation.

Network security has long been faced with the problem of the lack of real world network traffic data verification, the lack of effective collection of real traffic data and threat intelligence mechanism. However, maintaining a real world traffic data is expensive and collecting real world traffic is difficult in practice. Based on the above considerations, the proposed framework adopts two strategies: the first is the construction of a serverless computing platform, and the second is the collection of data using a two-sided marketing strategy. To connect two different user groups in a network through a two-way marketing strategy. A user can upload encrypted traffic data through the provided web interface to obtain security incident reports; another category of users can contribute threat intelligence to the provided platform through MISP and TheHive's collaborative analysis mechanism to obtain more relevant threat intelligence based on encrypted traffic data.

The contributions of this paper are as follows. First, this paper presents a two-sided market service model for malicious traffic collection. Second, our threat detection mechanism is based on scalable MapReduce algorithm and is constructed in a server-less architecture. Finally, we provide a web site: <https://netflowtotal.firebaseio.com/> for proof of concept.

2. RELATED WORK

Cyber threat information includes indicators of compromise; tactics, techniques, and procedures used by threat actors; suggested actions to detect, contain, or prevent attacks; and the

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findings from the analyses of incidents [2]. CTI is what cyber threat information becomes once it has been collected, evaluated in the context of its source and reliability, and analyzed through rigorous and structured tradecraft techniques by those with substantive expertise and access to all-source information [3]. If organizations can share cyber threat information, such as the detected malicious samples, it will be helpful to prevent the continuous expansion of security incidents in the shortest time. To encourage greater sharing of cyber threat information among organizations, NIST (National Institute of Standards and Technology) has produced a guide called SP 800-150. As cyber threat intelligence is shared, the specifications and protocol for sharing have evolved, including CybOX (Cyber Observable eXpression), OpenIOC, YARA, STIX (Structured Threat Information eXpression) and TAXII (Trusted Automated eXchange of Indicator Information). If the organization has the ability, it can build a Threat Intelligence Platform (TIP) by itself. Otherwise, it can only apply for Threat Intelligence Services (TIS) from security vendors. A threat intelligence platform for gathering, sharing, storing and correlating Indicators of Compromise of targeted attacks, threat intelligence, financial fraud information, vulnerability information or even counter-terrorism information [4]. Many studies have shown the importance of mining the unknown cyber threat information within an organization [5-7,19-20]. Past researches have focused on building systems at corporate gateways to collect network traffic and machine learning analysis to extract potentially malicious domains [5-7]. However, not all organizations have the ability to build systems that are compatible with big data collection and machine learning analytics. Therefore, this paper proposes and implements a serverless platform to solve

this dilemma.

Serverless computing was popularized by AWS (Amazon Web Services) Lambda in 2014 [8]. Other vendors followed in 2016 with the introduction of Google Cloud Functions [9], Microsoft Azure Functions [10] and IBM OpenWhisk [11]. Serverless computing is emerging as a new and compelling paradigm for the deployment of cloud applications, largely due to the recent shift of enterprise application architectures to containers and microservices [12]. Serverless computing refers to the cloud computing execution model, in which the cloud provider runs the server and dynamically manages the allocation of machine resources. It provides development environment, allocation and reallocation of resources when needed, storage and networking facility virtually. It satisfies the on-demand needs of the user. It facilitates the sharable resources “as-a-service” model. Serverless computing simplifies the process of deploying code to a production environment. The advantages of serverless computing are lower costs, simplified scalability, simplified backend code and quicker turnaround [13].

The core capability of a serverless platform is that of an event processing system. The service must manage a set of user-defined functions, take an event sent over HTTP or received from an event source, send the event to the function instance, wait for a response, gather execution logs, make the response available to the user, and stop the function when it is no longer needed. The challenge is to implement such functionality while considering metrics such as cost, scalability, and fault tolerance. The serverless platform presented in this paper is described in the next section

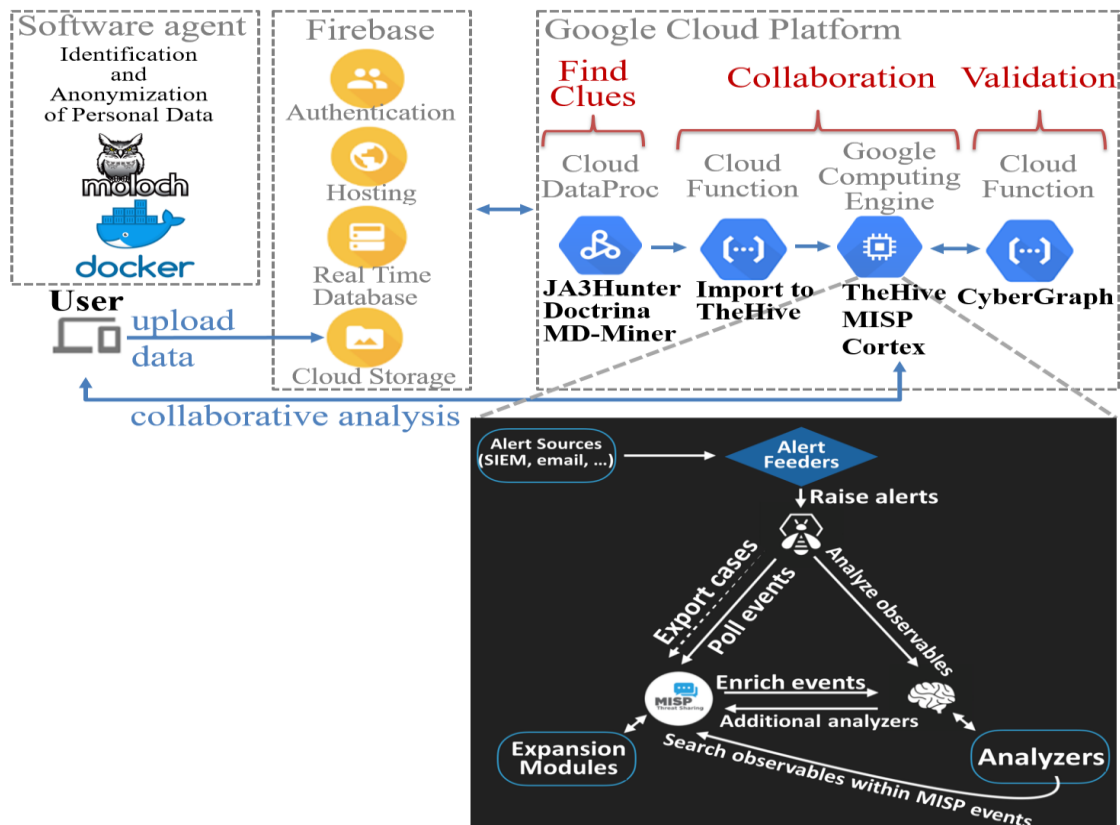


Figure 1. Framework of NetFlowTotal.

3. ARCHITECTURE

The holistic framework of the approach proposed in this paper is shown in Figure 1. This framework is divided into two main parts, one is Web Application and the other is event processing pipeline. The Web Application part is a serverless architecture built by Firebase. Firebase is a Backend-as-a-Service (BaaS) app development platform that provides hosted backend services such as a real time database, cloud storage, authentication, crash reporting, remote configuration, and hosting for user's static files. This web application handles the user interface and event source. The source of the event is the log file uploaded by the user. The user records the network traffic through the packet capture program side into a PCAP file, and extracts the log file from the PCAP file. At present, the log files of three protocols are mainly analyzed namely HTTP log, pDNS (Passive DNS) log and JA3 log. Once the user uploads the log file, it will be stored in Google cloud storage, and the subsequent event processing pipeline will be completed through the micro services based on Google Cloud Platform (GCP) framework.

Another part named event processing pipeline is composed of four micro services. Micro services are smaller segments of an application that run independently of each other. Each service is independently deployable on a potentially different platform and technological stack. It can run in its own process while communicating through lightweight mechanisms such as RESTful or RPC-based APIs. The event processing pipeline can be divided into three parts of micro-services: (1) Find Clues, (2) Collaboration, (3) Validation.

NetFlowTotal provides three kind find clues micro-services, which are long-day analysis of HTTP log, pDNS log and JA3 log respectively. HTTP log analysis is based on the MD-Miner algorithm [6]; and pDNS logs were analyzed using the Doctrina algorithm [7]; the analysis of JA3 logs is based on JA3Hunter's algorithm [14]. Above algorithms are constructed in the framework of MapReduce to meet the scalability requirements of huge amount of data. This paper turned above algorithms into serverless micro-service using the Cloud Dataproc WorkflowTemplates API. The Cloud Dataproc WorkflowTemplates API provides a flexible and easy-to-use mechanism for managing and executing workflows. It creates the Hadoop cluster, runs the jobs, stores the analysis results in the Google cloud storage, and then deletes the Hadoop cluster when the jobs are finished.

The collaborative micro-services are mainly divided into two parts: The first part is to transfer the analytical results of Find Clues micro-services from Google storage to The Hive through the Google Cloud Function. Cloud Function is a Function as a Service (FaaS) product that provides a platform allowing customers to develop, run, and manage application functionalities without the complexity of building and maintaining the infrastructure typically associated with developing and launching an app. Use cases for FaaS are associated with "on-demand" functionality that enables the supporting infrastructure to be powered down and not incurring charges when not in use. The second part is TheHive's services. TheHive is a scalable 3-in-1 open source and free Security Incident Response Platform, tightly integrated with MISP (malicious software information sharing platform), designed to information security practitioners dealing with security incidents that require rapid investigation and processing. The purpose of TheHive services under the proposed framework is to provide a collaborative analysis mechanism to attract users to share threat information and achieve a positive cycle of network effects in two-way marketing strategy. We build the three open sources: TheHive, MISP and Cortex on the cloud through Google Compute Engine services. Google Compute Engine (GCE) is the Infrastructure as a Service (IaaS) component of Google Cloud Platform.

In the part of the Validation micro-service, we implement CyberGraph algorithm through Google cloud function. CyberGraph is a threat intelligence verification mechanism in event processing pipeline, mainly through external information to verify the credibility of the Observables (IOC). The more detail describes in [14]. CyberGraph can be integrated as an analyzer of Cortex. Observables, such as IP and email addresses, URLs, domain names, files or hashes, can be analyzed one by one or in bulk mode using a Web interface through Cortex.

4. USE CASE

In the find clues step, the HTTP, HTTPS and pDNS log about network traffic can be extracted easily by the converter, respectively. These logs will be applied to MD-Minder, Doctrina and JA3Hunter system for discovering the hidden network threats as shown in Figure 2. The threats will be built into TheHive in case of suspicious hosts (Machine) and IOC (Observable). In the validation step, MISP and CyberGraph use external information to assist in the credibility of the IOC(Observables) identification. Finally, the sharing functionality of the MISP can ease data sharing using a different model of distributions.

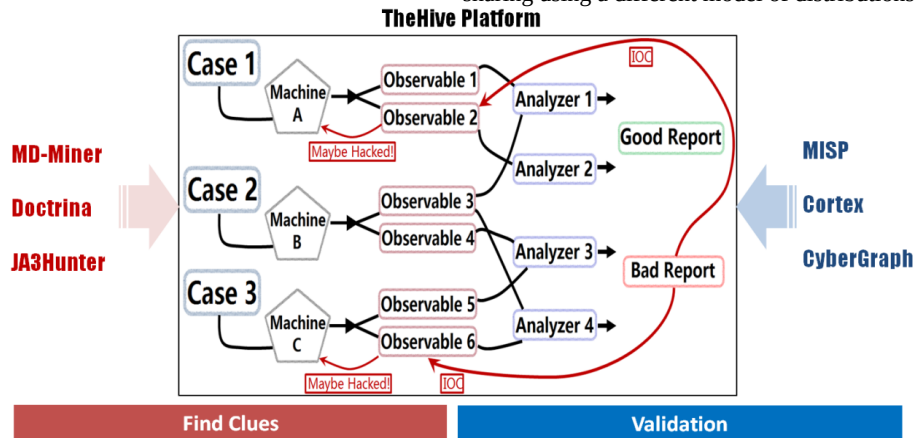


Figure 2. Operation flow of NetFlowTotal.

4.1 Find Clues

As shown in Figure 3, NetFlowTotal has three mechanisms for finding clues of malicious threats against traffic: MD-Miner, Doctrina, and JA3hunter. The user converts net traffic into HTTP logs, pDNS logs and JA3 logs through NetFlowTotal extraction program (netboxagent). And then, user can upload HTTP logs to the MD-Minder system, adopt pDNS logs to the Doctrina system, and use JA3 logs to take advantage of the JA3Hunter system to dig out hidden network threats. Among of these three systems, the MD-Minder system and Doctrina system combine annotated bipartite graph with supervised machine learning, and JA3Hunter uses BF-IDF to generate weights for different JA3 fingerprints to obtain suspicious encrypted JA3 fingerprints. These three systems can know which internal hosts communicate with the malicious network destinations (e.g., malicious domains and IPs), which is helpful for the digital forensic.

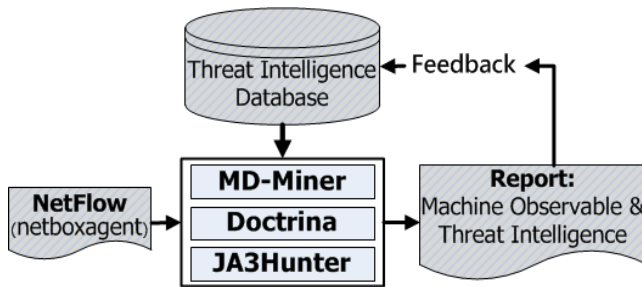


Figure 3. The process of NetFlowTotal.

4.1.1 MD-Miner (HTTP log)

MD-Miner monitors the network traffic to build an annotated bipartite graph named process-domain graph representing who is connecting what [6]. As shown on Figure 4, the process nodes indicate the unique user-agent and client-IP pair in the HTTP logs. That means used the user-agent (UA) + client-IP as a characteristic to distinguish the distinct process. Then, by applying the feature extraction method and machine learning modeling detailed in [6], MD-Miner can find new unknown domains that are likely to be malicious.

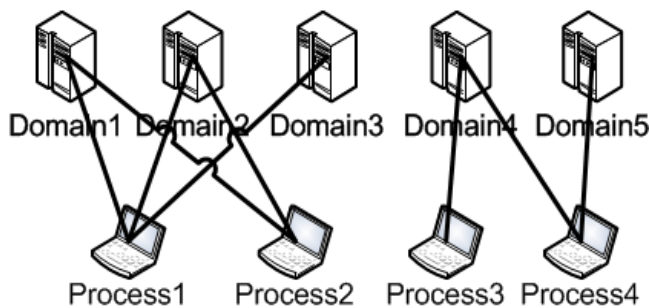


Figure 4. MD-Miner's process-domain graph.

4.1.2 Doctrina (pDNS log)

The main idea of the Doctrina system builds an annotated bipartite graph named machine-domain graph for representing that who is querying what relationship to discover the new malware-control domains. As shown on Figure 5, nodes represent either machines or domain names, and an edge between a machine and a domain represent that the machine queried the domain during the considered traffic observation time window. After machine-

domain graph constructed, Doctrina leverage the attributes in the pDNS logs to extract Machine Behavior (3 features), Domain Activity (4 features), IP abuse (4 features), BlackAnswer Profile (3 features), BlackSubnet Profile (3 features) and BlackTTL Profile (3 features). Once the features of each domain are extracted, Doctrina use the parallel processing power of spark to build random forest based decision tree model which is applied to the problem of malware-control domain detection [7].

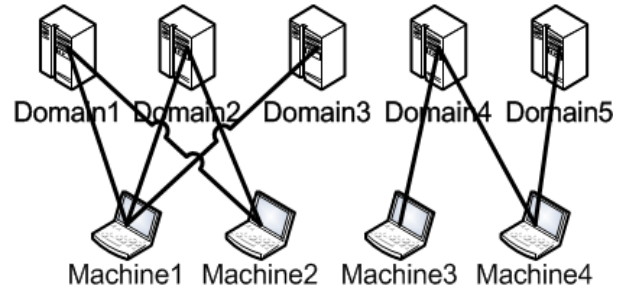


Figure 5. Doctrina's machine-domain graph.

4.1.3 JA3Hunter (HTTPS log)

JA3Hunter is a system for analyzing encrypted network traffic by using SSL/TLS client fingerprint and threat intelligence techniques. Through the IP address of the blacklist and the SSL/TLS client fingerprint connected to the blacklist, JA3Hunter proposes a BF-IDF (Term Frequency-Inverse Document Frequency) algorithm, which can extract the suspicious SSL/TLS client fingerprint. Consequently, JA3Hunter can find hidden malicious IP which cannot be recognized by other external threat information providers from encrypted traffic [14].

4.2 Collaboration Analysis (TheHive, MISP, Cortex)

We use an open source and free security incident response platform TheHive [15] and Cortex [16] to analysis the new observables (such as IP addresses, domain names, file hashes, files, and URLs) which are generated from MD-Miner, Doctrina, and JA3Hunter. Security researchers and analysts can collaborate on investigations simultaneously with this platform. After investigations and analysis were finished, new indicators of compromise (IOCs) and related threat intelligence could be export to an open source threat intelligence platform MISP (Malware Information Sharing Platform) by which collaborative knowledge can be stored, correlated, shared to/from different sources and communities. Basically the integrated analysis process with NetFlowTotal platform can be divided into following 6 steps:

- 1). Users upload new network data to NetFlowTotal platform.
- 2). NetFlowTotal generate new TheHive cases with suspicious associated observables automatically.
- 3). Observables were sent to Cortex analyzers for further analyzing.
- 4). Security researchers and analysts collaborate to investigate new IOCs and threats base on the analysis reports from Cortex.
- 5). Share new found IOCs to MISP
- 6). Repeat step 1 to 5.

4.3 Validation (CyberGraph)

To further verify the observable, the VirusTotal will be used to analyze these IP addresses and domains. Note that, VirusTotal simply aggregates the output of different antivirus vendors and URL scanners; it does not provide a standard answer of malicious IP or domain. To further analyze the other results which are not

detected as malicious site, the external threat intelligence of these IP addresses or domains will be collected, such as passive DNS records, file communicated records, and use these information to construct the relation graph of the IP or domain.

5. EVALUATION

To evaluate our find clue mechanisms effectiveness, we deployed MD-Miner, Doctrina and JA3Hunter in one real world enterprise network environment, which is located in Taiwan. We collected the network traffic over a period of two months (from April 1, 2019 to May 31, 2019). HTTP logs, pDNS logs and JA3 logs extracted from this period of traffic are called ENT_{N1} , ENT_{N2} and ENT_{N3} respectively. The details of the collected data are shown in Table 1, Table 2 and Table 3.

Table 1. The Attributes Of HTTP Log in ENT_{N1}

Size	# of ClientIPs	# of Domains	# of HTTP requests
5 G	1,849	1,094,112	23,914,608

Table 2. The Attributes Of pDNS Log in ENT_{N2}

Size	# of ClientIPs	# of Domains	# of passive DNS records
14.3 G	291	4,021,505	105,335,597

Table 3. The Attributes Of JA3 Log in ENT_{N3}

Size	# of ClientIPs	# of JA3 Fingerprints	# of Logs
19.2 G	1757	911	86,450,474

5.1 Accuracy Evaluation

MD-Miner and Doctrina evaluate their performance by receiver operating characteristic curves (ROC curves) and long-term network traffic in the real world. The ROC curve shows true positive rate (TPR) on Y-axis which means the ratio of all actual positive samples that were correctly classified to be positive, against false positive rate (FPR) on X-axis which means the ratio of samples that were actual negative was classified to be false positive. In order to obtain TPR and FPR , we have to label all known samples as negative (n) or positive (p) that also represent as benign and black. After apply the labeled samples into evaluated system (MD-Miner or Doctrina), we would receive the outcomes and get four values for calculate TPR and FPR , the first value is true positive (TP) means outcome from a prediction is p and the actual value is also p ; the second value is false positive (FP) means outcome from a prediction is p but the actual value is n ; the third value is true negative (TN) means outcome from a prediction is n and the actual value is also n ; and the fourth value is false negative (FN) means outcome from a prediction is n but the actual value is p . At last, we could consider, $TPR = TP / (TP + FN)$, $FPR = FP / (FP + TN)$ to carry out ROC curve and obtain area under the curve (AUC) of ROC which is equal to the probability that a classifier will rank a randomly chosen positive instance higher than a randomly chosen negative one (assuming 'positive' ranks higher than 'negative') [17]. In addition, this experiment would conduct 10-fold cross validation on these three feature vectors respectively to verify their effectiveness. In 10-fold cross-validation, the original sample is randomly partitioned into ten equal sized subsamples. One of the separate subsample is retained as the data for the validation model, and the other nine samples are used for training. Cross-validation is repeated ten times and each subsample is validated once. The average of ten times results is eventually resulting in a single estimate. The advantage of this method is that repeated random

sub-sampling is used for both training and validation, and each result is used for validation exactly once. 10-fold cross validation is the most commonly used method for testing accuracy [18].

The results showed that the 0.97 AUC could be achieved on the ENT_{N1} dataset by MD-Miner and the 0.98 AUC by Doctrina on the ENT_{N2} dataset, which was consistent with the results of the [6] and [7] experiments. Experiments have shown that MD-Miner and Doctrina are feasible methods for malware-control domain detection.

5.2 Collaboration and Validation Analysis

The network traffic of ground truth will be used to evaluate and verify the effectiveness of the proposed system. The study collects a total of 19.2 GB of enterprise network traffic for ENT_{N3} . JA3Hunter can detect the observable for collaboration analysis as shown on Figure 6. The observable 216[.]239[.]34[.]21 is potentially a new malicious IP address in this case. It should be marked as IOCs and share to MISP as well. And JA3 fingerprints 14eff09463c9d6dc6e14e954eba239c2 from JA3Hunter matched a known threat in MISP.



Figure 6. A sample case of collaborative analysis with JA3Hunter

For further analyzing the destination IP address 216[.]239[.]34[.]21 is potentially a new malicious IP address, we collect the external threat intelligence of the IP, such as the passive DNS records and the file communicated records, and use this information to construct a relationship graph of a domain. Given the situation in the relationship graph of the IP, we observe that the detected IP is directly or indirectly communicated with the malicious files as shown in Figure 7. The IP address 216[.]239[.]34[.]21 communicates to the files, and these files are detected for malicious. And then, the IP has been hosted to dsdownload[.]pushalert[.]co, *[.]surat4u[.]com and *[.]bunchchiropractic[.]net. Therefore, the external threat intelligence shows that these domains have been resolved to 216[.]239[.]38[.]21, 216[.]239[.]36[.]21 and 216[.]239[.]32[.]21, and these IP addresses were communicated with the malicious files. A series of outward relationship, the IP address 216[.]239[.]34[.]21 can be judged to be the malicious IP address. Thus, we can show JA3Hunter could detect the malicious IP address which was not be identified by other reputation service.

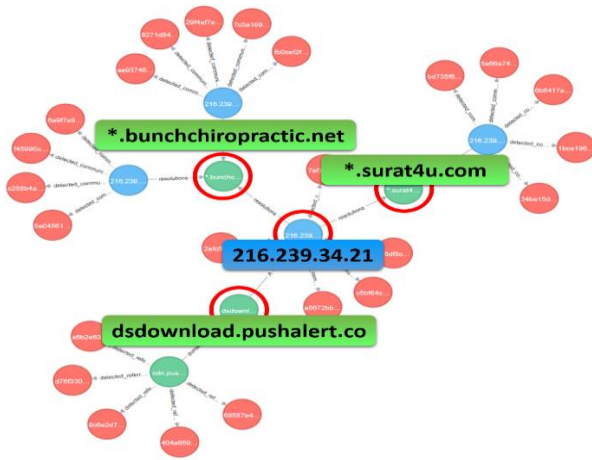


Figure 7. An example of IP relationship graph constructed by external threat intelligence.

5.3 Cost/Performance Evaluation (GCP cost, MapReduce scalability)

To verify the scalability and cost of the system, we tested execution time and cost for MD-Miner, Doctrina and JA3Hunter under cloud Dataproc clusters of different specifications. MD-Miner used the ENT_{N1} dataset; Doctrina uses the ENT_{N2} dataset; JA3Hunter uses the ENT_{N3} dataset. Table 5 lists the specifications, execution time, and cost of the cluster for MD-Miner, Doctrina and JA3Hunter respectively.

As you can see from table 5, the execution time of a task decreases linearly as the cluster size increases. Experimental results show that MD-Miner, Doctrina and JA3Hunter are scalable under the MapReduce framework.

Table 4. Cost Analysis of Find Clue Mechanism

Cluster specification	MD-Miner for ENT_{N1}		Doctrina for ENT_{N2}		JA3Hunter for ENT_{N3}	
	run time	cost	run time	cost	run time	cost
1 master 4 nodes n1-stand-4 YARN Core: 16 YARN Memory: 48 GB	340 SEC	0.18 USD	1759 SEC	0.93 USD	1202 SEC	0.63 USD
1 master 9 nodes n1-stand-4 YARN Core: 36 YARN Memory: 108 GB	188 SEC	0.21 USD	1160 SEC	1.27 USD	736 SEC	0.82 USD
1 master 14 nodes n1-stand-4 YARN Core: 56 YARN Memory: 168 GB	154 SEC	0.26 USD	956 SEC	1.35 USD	560 SEC	0.94 USD

In the execution cost part, the Find Clues mechanism uses the cloud Dataproc workflow template, so the cluster is only built during the run and deleted after the run. As shown in table 4, using a cluster twice the size reduces time by about a third but increases relative costs. Cloud Dataproc workflow templates provide users with a fairly flexible choice based on their needs.

6. DISCUSSION AND CONCLUSION

Our project will help the cyber security communities to share net flow data and threat intelligence. By facilitating the collection of net flow data and threat intelligence in the GCP ecosystem and combined with our network threat analysis technology, we will build a global network threat intelligence system and a gold standard evaluation benchmark for machine learning on network security. This research will pave the way for major research universities and machine learning solutions in network security.

7. REFERENCES

- [1] Brown, R. and Lee, R. M. 2019. *The Evolution of Cyber Threat Intelligence (CTI): 2019 SANS CTI Survey*. Technical Report. SANS Institute.
- [2] Chris, J., Lee, B., David, W., Julie, S. and Clem, S. 2016. *Guide to Cyber Threat Information Sharing*. Technical Report. National Institute of Standards and Technology.
- [3] Intel and Analysis Working Group. 2019. What is Cyber Threat Intelligence?, Center for Internet Security. Retrieved from <https://www.cisecurity.org/blog/what-is-cyber-threat-intelligence/>
- [4] MISP - Open Source Threat Intelligence Platform and Open Standards for Threat Information Sharing. 2019. Retrieved from <https://www.misp-project.org/>
- [5] Jeng, T. H., Chen, Y. M., Chen, C. C., Huang, H. C. and Chou, K. S. 2018. CC-Tracker: Interaction Profiling Bipartite Graph Mining for Malicious Network Activity Detection. In *Proceedings of the IEEE Conference on Dependable and Secure Computing*, (Kaohsiung, Taiwan, Dec. 10 - 13, 2018). 323-330. DOI= <https://doi.org/10.1109/DESEC.2018.8625170>.
- [6] Sun, J. H., Jeng, T. H., Chen, C. C., Huang, H. C. and Chou, K. S. 2017. MD-Miner: Behavior-Based Tracking of Network Traffic for Malware-Control Domain Detection. In *Proceedings of the IEEE International Conference on Big Data Computing Service and Applications*, (San Francisco, USA, April 06 - 09, 2017). 96-105. DOI= <https://doi.org/10.1109/BigDataService.2017.16>.
- [7] Chen, S. Y., Jeng, T. H., Huang, C. C., Chen, C. C. and Chou, K. S. 2017. Doctrina: Annotated Bipartite Graph Mining for Malware-Control Domain Detection. In *Proceedings of the 7th International Conference on Communication and Network Security*. (Tokyo, Japan, Nov. 24 - 26, 2017). 67-75. DOI= <https://doi.org/10.1145/3163058.3163061>.
- [8] Wagner, T. 2014. Getting started with AWS lambda. *Aws re:invent 2014 (MBL202)*. Retrieved from <https://www.youtube.com/watch?v=UFj27laTWQA>
- [9] Google Cloud. 2019. Google Cloud Functions. Retrieved from <https://cloud.google.com/functions/>
- [10] Microsoft Azure. 2019. Azure Functions. Retrieved from <https://functions.azure.com/>
- [11] Incubator-openwhisk. 2019. Openwhisk. Retrieved from <https://github.com/openwhisk/openwhisk>

- [12] Baldini, I., Castro, P., Chang, K., Cheng, P., Fink, S., Ishakian, V., Mitchell, N., Muthusamy, V., Rabbah, R., Slominski, A. and Suter, P. 2017. Serverless Computing: Current Trends and Open Problems. *Research Advances in Cloud Computing*. Springer, 1-20.
- [13] Cloudflare. 2019. What is serverless computing?. Retrieved from <https://www.cloudflare.com/learning/serverless/what-is-serverless/>
- [14] Jeng, T. H., Luo, W. Y., Huang, C. C., Chen, C. C., Chang, K. H. and Chen, Y. M. 2019. Cloud Computing for Malicious Encrypted Traffic Analysis and Collaboration. *International Journal of Grid and High Performance Computing*. In press.
- [15] TheHive. 2019. Security Incident Response Platform. Retrieved from <https://thehive-project.org/>
- [16] Cortex. 2019. Powerful Observable Analysis and Active Response Engine. Retrieved from <https://github.com/TheHive-Project/Cortex>
- [17] Hastie, T. Tibshirani, R. and Friedman, J. 2009. The Elements of Statistical Learning: Data Mining, Inference, and Prediction. Springer. New York.
- [18] Cooke, E. Jahanian, F. and McPherson D. 2005. The Zombie Roundup: Understanding, Detecting, and Disrupting Botnets. In *Proceedings of the Workshop on Steps to Reducing Unwanted Traffic on the Internet*. (Cambridge, MA, July 7, 2005). 39-44.
- [19] Hong, K. F., Chen, C. C., Chiu, T. T. and Chou, K. S. 2015. Ctracer: Uncover C&C in Advanced Persistent Threats Based on Scalable Framework for Enterprise Log Data. In *Proceedings of the IEEE International Congress on Big Data*. (New York, USA, June 27 - July 2, 2015). 551-558. DOI=<https://doi.org/10.1109/BigDataCongress.2015.86>.
- [20] Hong, K. F., Chen, C. C., Chiu, T. T. and Chou, K. S. 2015. Scalable Command and Control Detection in Log Data through UF-ICF Analysis. In *Proceedings of the International Carnahan Conference on Security Technology*. (Taipei, Taiwan, Sept. 21 - 24, 2015). 293-298. DOI=<https://doi.org/10.1109/CCST.2015.7389699>.